

```
from google.colab import files
uploaded = files.upload()
```

Chọn tệp lab4_word_embeddings_pack.zip

- **lab4_word_embeddings_pack.zip**(application/x-zip-compressed) - 4250 bytes, last modified: 16/10/2025 - 100% done

Saving lab4_word_embeddings_pack.zip to lab4_word_embeddings_pack.zip

```
!unzip lab4_word_embeddings_pack.zip -d lab4_word_embeddings_pack
%cd lab4_word_embeddings_pack
```

```
Archive: lab4_word_embeddings_pack.zip
  creating: lab4_word_embeddings_pack/src/
  creating: lab4_word_embeddings_pack/test/
  creating: lab4_word_embeddings_pack/results/
  creating: lab4_word_embeddings_pack/data/
 inflating: lab4_word_embeddings_pack/requirements.txt
 inflating: lab4_word_embeddings_pack/README.md
  creating: lab4_word_embeddings_pack/src/representations/
 inflating: lab4_word_embeddings_pack/src/representations/__init__.py
 inflating: lab4_word_embeddings_pack/src/representations/word_embedder.py
 inflating: lab4_word_embeddings_pack/test/lab4_test.py
 inflating: lab4_word_embeddings_pack/test/lab4_embedding_training_demo.py
 inflating: lab4_word_embeddings_pack/test/lab4_spark_word2vec_demo.py
/content/lab4_word_embeddings_pack
```

```
!pip install -q gensim==4.3.3 scikit-learn==1.5.2 pyspark==3.5.1
```

```
!ls
```

```
data README.md requirements.txt results src test
```

```
!touch test/__init__.py
```

```
!python -m test.lab4_test
```

```
[=====] 100.0% 66.0/66.0MB downloaded
```

```
Vector('king') shape: (50,)
similarity('king', 'queen') = 0.7839043818384214
similarity('king', 'man')   = 0.5309376965998382
```

```
Top-10 most similar to 'computer':
```

1. computers	0.9165
2. software	0.8815
3. technology	0.8526
4. electronic	0.8126
5. internet	0.8060
6. computing	0.8026
7. devices	0.8016
8. digital	0.7992
9. applications	0.7913
10. pc	0.7883

```
Doc embedding for: The queen rules the country.
```

```
shape: (50,)
head10: [ 0.0244  0.378 -0.6382  0.0128  0.0524  0.1195 -0.3165 -0.0878  0.0776
 -0.5418]
```

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Chọn tệp Chưa có tệp nào được chọn Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving c4-train.00000-of-01024-30K-ison.gz to c4-train.00000-of-01024-30K-ison.gz

```
!python -m test.lab4_embedding_training_demo
```

Top-5 similar to 'computer': [('com', 0.9982714056968689), ('place', 0.9982286691665649), ('enjoy', 0.998176097869873), ('etc', 0.998176097869873), ('king', 0.998176097869873)]
Analogy king - man + woman: [('europe', 0.9894932508468628), ('expected', 0.9894914627075195), ('parents', 0.9894763827323914)]

```
!python -m test.lab4_spark_word2vec_demo
```

Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
25/10/16 16:57:46 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

word	similarity
198x	0.6599948406219482
desktop	0.6528733968734741
software	0.6435503959655762
computers	0.6379222273826599
programming	0.6299683451652527